1.5 - Scaling to 100+ Correlated Bonds: the Modulus Matrix and its Factor-Reduction Error Bound

REFLEX project - `math-theory`

STATUS: DONE (Priority 5, derivation/theorem). This document discharges the *mathematical* content of methodology target **1.5**. It lifts the scalar boundary of 1.1 from one bond to a universe of d correlated bonds, replacing the scalar modulus $m = \varepsilon\beta/\gamma$ by a $d \times d$ **modulus matrix**

$$M = \beta\,\Gamma^{-1}E, \qquad \text{stable} \iff \rho(M) < 1,$$

and makes the $d = 100+$ regime tractable and honest via the factor structure already present in `bonds.py`: Γ is diagonal-plus-low-rank, so M and its boundary are computable in $O(dk^2)$ (Woodbury / Bergault-Guéant), and truncating to k factors perturbs the boundary by an amount linear in the residual factor variance $\|R\| = \lambda_{k+1}(C)$ (Theorem 1).

One-line thesis. One bond is stable iff a scalar $m < 1$. A book of d correlated bonds is stable iff the $d \times d$ matrix $M = \beta\Gamma^{-1}E$ has spectral radius below 1 - and because inventory risk couples bonds through their shared factor covariance, the most fragile direction is the **market factor**: the whole book re-quoting in unison. The curse of dimensionality is defused by the same factor structure that causes it.

1 Carry-over and new notation

For each bond i (own microstructure $A_i, k_i, c_{t,i}, I_i$):

$$\begin{aligned}
\gamma_i &= P[\,2w + A_i\rho k_i e^{-k_i h_i}(2 - k_i h_i) + \lambda_{q,i}\,], & \beta &= P, \\
\varepsilon_i &= \rho\bar{g}\,\alpha f I_i c_{t,i} e^{-c_{t,i}h_i}, & m_1^{(i)} &= \varepsilon_i\beta/\gamma_i.
\end{aligned}$$

Symbol	Reading	Definition / source
d	number of bonds	`bonds.n_bonds`
$\boldsymbol{h}$	spread vector	per-bond b_h
C	$d \times d$ correlation matrix	`BondUniverse.corr`
Σ	covariance $= D_\sigma C D_\sigma$	$D_\sigma = \mathrm{diag}(\sigma_i)$
$B, \Lambda, D_{\mathrm{idio}}$	loadings / factor cov / idiosyncratic	$C = B\Lambda B^\top + D_{\mathrm{idio}}$
k	retained factors, $k \ll d$	$1 +$ `n_sectors`
ζ	inventory risk aversion	$\propto$ `reward.inv_risk_weight`
G	$\mathrm{diag}(dq_i/dh_i)$	§2
Γ	Hessian $= D_\gamma + \zeta' G\Sigma G$	§2
E	$\mathrm{diag}(\varepsilon_1..\varepsilon_d)$	§2
M	modulus matrix $= \beta\Gamma^{-1}E$; stable iff $\rho(M) < 1$	§2–§3
R	dropped-factor residual $C - C_k$; $\lVert R\rVert = \lambda_{k+1}(C)$	§5

2 Setup: the d-bond maker and the `bonds.py` factor model

A single dealer quotes d bonds, holding inventory $\boldsymbol{q} \in \mathbb{R}^d$; the new ingredient is cross-bond inventory risk through Σ.

Assumption 1 (factor covariance is `bonds.py`'s)**.** *$C =$ `BondUniverse.corr` is, by construction, a factor model: a global market factor (loading $\sqrt{\texttt{global\_factor}}$ on every bond) + per-sector blocks + idiosyncratic residual, i.e. $C = B\Lambda B^\top + D_{\mathrm{idio}}$ with $k = 1 +$ `n_sectors` factors. $\Sigma = D_\sigma C D_\sigma$.*

Assumption 2 (per-bond toxic channel; quadratic inventory value)**.** *Toxic flow is per-bond (ε_i); the cross-sectional coupling is the inventory penalty $(P\zeta/2)\boldsymbol{q}^\top\Sigma\boldsymbol{q}$ (Guéant/Avellaneda-Stoikov form).*

3 The multi-bond objective and the modulus matrix

Summing 1.1 termwise and subtracting the inventory penalty (with $G = \mathrm{diag}(dq_i/dh_i)$):

$$J(\boldsymbol{h}) = \sum_i P\big[h_i A_i e^{-k_i h_i}\rho + h_i T_i - \psi_i T_i - w(h_i - h_{\mathrm{ref}})^2\big] - \tfrac{P\zeta}{2}\boldsymbol{q}(\boldsymbol{h})^\top \Sigma\, \boldsymbol{q}(\boldsymbol{h}).$$

Differentiating in $\boldsymbol{h}$:

$$\Gamma := -\partial^2_{\boldsymbol{hh}} J = D_\gamma + \zeta' G\Sigma G, \quad \zeta' := P\zeta, \quad D_\gamma = \mathrm{diag}(\gamma_i^0),$$
$$\partial^2 J/\partial h_i \partial T_j = P\,[i{=}j] \;\Rightarrow\; \beta I,\; \beta = P; \qquad d\tau_i/dh_i^{\mathrm{dep}} = -\varepsilon_i \;\Rightarrow\; E = \mathrm{diag}(\varepsilon_i).$$

$\Gamma \succ 0$ with off-diagonals $\zeta' g_i \Sigma_{ij} g_j$ - the cross-bond coupling. The FOC IFT ($F_h = -\Gamma$, $F_T = \beta I$, $d\tau/d\boldsymbol{h}^{\mathrm{dep}} = -E$) gives $J_{\mathrm{BR}} = -\beta\Gamma^{-1}E$, so the loop contracts iff $\rho(M) < 1$ with

$$\boxed{M = \beta\Gamma^{-1}E = P\,\Gamma^{-1}\mathrm{diag}(\varepsilon_i)} \tag{1}$$

4 The boundary and its structure

Sanity checks. Uncorrelated book: Σ, Γ diagonal, $M = \mathrm{diag}(m_1^{(i)})$, $\rho(M) = \max_i m_1^{(i)}$ (the least-stable bond sets the boundary). $d = 1$: $\rho(M) = m_1$. ✓

Market-factor mode. The coupling in Γ lives entirely in the low-rank $\zeta' G(D_\sigma B)\Lambda(D_\sigma B)^\top G$, so the top eigenvector of M aligns with the top factor of Σ - the **global market factor**. The fragile direction is the whole book re-quoting together (synchronised correlated inventory, undiversifiable), the cross-sectional twin of 1.3's common dealer mode; idiosyncratic perturbations are damped more.

Concentration. At fixed per-bond feedback, a concentrated (high `global_factor`) universe reaches $\rho(M) = 1$ at a lower f than a diversified one (§7.1): correlation, not count, is the systemic variable.

5 Factor reduction: $O(dk^2)$ tractability

Fold the idiosyncratic part into the diagonal and invert by Woodbury:

$$\Gamma = D + U\Lambda U^\top, \quad D = D_\gamma + \zeta' G D_\sigma D_{\text{idio}} D_\sigma G, \quad U = \sqrt{\zeta'}\, G D_\sigma B \ (d \times k),$$
$$\Gamma^{-1} = D^{-1} - D^{-1}U(\Lambda^{-1} + U^\top D^{-1} U)^{-1} U^\top D^{-1}.$$

Forming Γ^{-1}, $M = P\Gamma^{-1}E$, and $\rho(M)$ by power iteration costs $O(dk^2 + k^3)$ instead of $O(d^3)$ - Bergault-Guéant's reduction applied to the *stability operator*. Each Mv is a diagonal solve plus a rank-k update; no dense $d \times d$ matrix is materialised.

6 The factor-truncation error bound

Retain the top k factors, $C \approx C_k = B_k \Lambda_k B_k^\top$, dropping $R := C - C_k$ with $\|R\|_2 = \lambda_{k+1}(C)$; let Γ_k, M_k use C_k.

Theorem 1 (dimensionality-reduction error). *With $\gamma_{\min} = \lambda_{\min}(D_\gamma) > 0$, $\varepsilon_{\max} = \max_i \varepsilon_i$, $g_{\max} = \max_i |g_i|$,*

$$\|M - M_k\|_2 \le \frac{P\,\varepsilon_{\max}\,\zeta' g_{\max}^2}{\gamma_{\min}^2}\,\|R\|_2, \qquad |\rho(M) - \rho(M_k)| \le \kappa(V)\,\|M - M_k\|_2 = O(\lambda_{k+1}(C)),$$

where $\kappa(V)$ is the eigenvector conditioning of M (Bauer-Fike; $= 1$ when M is normal). Hence the feedback-gain boundary f^* solving $\rho(M(f)) = 1$ has $|f_k^* - f^*| = O(\lambda_{k+1}(C))$, linear in the residual factor variance and vanishing as $k \to d$.

Proof. $\Gamma, \Gamma_k \succeq D_\gamma$ give $\|\Gamma^{-1}\|, \|\Gamma_k^{-1}\| \le 1/\gamma_{\min}$. The resolvent identity with $\Gamma - \Gamma_k = \zeta' G D_\sigma R D_\sigma G$ yields $\|\Gamma^{-1} - \Gamma_k^{-1}\| \le (\zeta' g_{\max}^2 \sigma_{\max}^2/\gamma_{\min}^2)\|R\|$; multiply by $\|\beta E\| = P\varepsilon_{\max}$. Bauer-Fike bounds the eigenvalue perturbation of the diagonalisable M by $\kappa(V)\|M - M_k\|$. Since E (hence M) is linear in f, $d\rho/df > 0$ is bounded below on the operating range, so dividing preserves the rate. □

This is the Bergault-Guéant (2021) Proposition 4 analogue: the reduction error is closed-form in the residual variance, so a k-factor phase diagram carries a rigorous $O(\lambda_{k+1}(C))$ band. Choosing k so $\lambda_{k+1}(C)$ (a computable eigenvalue of the known C) is below tolerance is the truncation rule.

7 Calibration

Real data: $\sigma_i \leftarrow \mathrm{DV01}_i\,\sigma_{\text{rate}}$ with $\mathrm{DV01}_i = \text{duration}_i \cdot \text{price}_i \cdot 10^{-4}$; k_i, A_i from MLE of $\lambda(\delta) = A_i e^{-k_i\delta}$ on inter-trade times; C from empirical bond-return correlations (rate/credit/sector PCA $\to B, \Lambda$); ζ from risk aversion. Then γ_i, ε_i from 1.1. **Synthetic (no data):** calibrate ε_i from the Cont-Kukanov-Stoikov informed-intensity slope $d\lambda_{\text{informed}}/d\delta$ (1.1 §8's third estimator), disclosed as synthetic.

8 Corollaries

Corollary 1 (diversification is a stability lever)**.** *$f^*(book) \sim f^*(single)/\chi(C)$ with $\chi(C) \geq 1$ the top-factor amplification ($= 1$ for diagonal C, growing with `global_factor`). Adding uncorrelated names leaves $\rho(M) = \max_i m_1^{(i)}$ unchanged; factor-correlated names inflate it.*

Corollary 2 (compose with 1.3 / 1.4)**.** *For N dealers $\times$ d bonds the fragile eigenvector is $(\mathbf{1}_{dealers}) \otimes$ (global bond factor) and $\rho_{\text{sys}} \sim N_{\text{eff}}\chi(C)m_1$, so stable iff $\varepsilon < \gamma/(N_{\text{eff}}\chi(C)\beta)$. The robust version (1.4) certifies iff $\hat{\rho}(M) + \delta_n < 1$; the factor reduction makes the cross-seed re-estimation over 100+ bonds affordable.*

9 Validation protocol and remaining code task

Predict: factorise C; build M via Woodbury; predict $\rho(M)$, f^*, the market-factor eigenvector, and the $O(\lambda_{k+1}(C))$ truncation error. **Measure:** generalise `response_modulus.py` to a CRN finite difference along the top factor eigenvector v_1, $\hat{\rho}(M) = \|\text{BR}(\boldsymbol{h}^* + \delta v_1) - \text{BR}(\boldsymbol{h}^* - \delta v_1)\|/(2\delta)$. **Compare:** $\hat{\rho}(M)$ vs f over a $d = 100+$ universe with median+IQR over universe draws; overlay the predicted boundary, the `global_factor`-sweep diversification law, and the shrinking k-factor error band. **Code (out of scope):** `analysis/factor_reduction.py` (factorise, Woodbury, error bound); factor-mode probe; per-bond calibration from `bonds.features`. $d = 1$ / diagonal C reproduces 1.1.

10 Honest caveats

- The inventory coupling is the modelled cross-sectional channel; $\zeta \to 0$ decouples into 1.1 bond-by-bond ($\rho(M) \to \max_i m_1^{(i)}$).
- $G = \text{diag}(dq_i/dh_i)$ is a first-order operating-point sensitivity; large fills / non-zero skew add cross-bond and nonlinear terms.
- Bauer-Fike $\kappa(V) > 1$ for strongly non-normal (very heterogeneous) books; report it rather than assuming $\kappa(V) = 1$.
- Calibration provenance: absent TRACE, disclose the synthetic CKS route.
- Reference-state constants / cap slack as in 1.1 §9; the phase diagram is a band over reference states *and* universe draws.
- Single dealer here; the N-dealer $\times$ d-bond Kronecker spectrum is the 1.3 $\times$ 1.5 follow-on.

References

- P. Bergault, O. Guéant. *Size Matters for OTC Market Makers: General Results and Dimensionality Reduction Techniques.* Mathematical Finance, 2021. arXiv:1907.01225. - factor reduction (§4) and residual-variance error bound (Prop. 4; §5). (Lit. #9.)
- R. Cont, A. Kukanov, S. Stoikov. *The Price Impact of Order Book Events.* J. Fin. Econometrics, 2014. arXiv:1011.6402. - synthetic ε_i calibration (§6). (Lit. #18.)

- Avellaneda-Stoikov; Guéant-Lehalle-Fernández-Tapia (GLFT) - fill intensity and quadratic inventory value (§1–§2).
- Weyl's inequality and the Bauer-Fike theorem (§5); the Woodbury identity (§4).
- Builds on `01-analytic-stability-boundary`; composes with `03-multi-dealer-systemic-risk` and `04-robust-uncertainty` (§7).